A Privacy-Preserving Data Generation Framework in Federated Learning for Enhanced IoT System Attack Detection

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Abstract*—Federated learning lets Internet of Things (IoT) gateways train an intrusion detection system (IDS) without pooling raw traffic, but rare attack classes are scattered across gateways and model updates can still leak training records. Federated synthetic data generation is a natural remedy, yet deep generators trained with differentially private stochastic gradient descent (DP-SGD) lose much of their utility at practical privacy budgets. This paper presents PP-FedData, a framework that adapts marginal-based private synthesis to IoT gateways. Its generator, FedDP-Marginal, releases noisy count tables in three rounds, splits integer Skellam noise across the gateways, sums the tables with Flower's SecAgg+ protocol, and samples class-conditional packets from a Chow–Liu tree. A federated conditional variational autoencoder (CVAE) trained with FedAvg, DP-SGD and SecAgg+ serves as the deep baseline under identical partitions, budgets and evaluation. On the DoS/DDoS-MQTT-IoT dataset (six classes, five non-IID gateways, three seeds), an IDS trained only on FedDP-Marginal data reaches a test macro-F1 of 0.389, 0.408 and 0.429 at $\varepsilon = 1$, 5 and 10 ($\delta = 10^{-5}$), against 0.227–0.289 for the best CVAE configurations and 0.447–0.504 for classifiers trained on pooled real data, while transmitting 0.45–0.60 MB per run instead of 27–68 MB. Secure aggregation costs 1.34 times the bytes and no utility, and a membership inference attack on the released tables separates the release without DP (AUC 0.560) from the releases with DP (0.514–0.531). Synthetic data do not improve an IDS when real data can be pooled; their value lies where raw traffic must stay local. A requirement-driven rule maps trust, privacy and bandwidth requirements to a configuration.

***Index Terms*—Federated learning, differential privacy, secure aggregation, synthetic data, marginal-based synthesis, intrusion detection, MQTT, Internet of Things.**

I. Introduction

The Message Queuing Telemetry Transport (MQTT) protocol carries the publish–subscribe traffic of many Internet of Things (IoT) deployments, and its brokers are exposed to denial-of-service (DoS) and distributed DoS (DDoS) attacks that abuse the protocol itself. The DoS/DDoS-MQTT-IoT dataset, captured on a physical testbed, records normal MQTT traffic together with ten such attack scenarios [1]. Intrusion detection systems (IDSs) based on machine learning learn these attacks from labeled traffic, so their quality depends on how much representative traffic of each attack they see. In practice that traffic is spread over many gateways that cannot pool it. Federated learning (FL) trains a shared model while the data stay local: a server distributes a global model and aggregates the clients' updates over several rounds [2]. Keeping data local, however, does not remove the privacy risk, because shared gradients can reveal training records [3], and privacy-preserving FL for IoT intrusion detection remains an active field [4].

Two difficulties combine in this setting. First, rare attack classes are scattered unevenly across gateways (non-IID data): a gateway may hold a handful of packets of a rare attack, or none, which biases any local detector toward the majority classes. Second, the two standard protections answer different threats. Secure aggregation (SecAgg) hides each client's update from the server and reveals only their sum [5], [6], but it says nothing about what the sum, the trained model or its synthetic data reveal. Differential privacy (DP) bounds that disclosure for any single record [7], at a cost in utility. Federated generative models have been proposed to synthesize the missing attack traffic [8]–[11]. These are deep networks: they exchange full models over tens of rounds and, when trained with DP stochastic gradient descent (DP-SGD) [12], spread the privacy noise over thousands of optimization steps.

Marginal-based synthesis takes another route: it measures low-order count tables with DP noise and samples records from a graphical model fitted to them [13]–[15]. Count tables are sums over records, which is exactly what secure aggregation computes. Discrete noise such as the Skellam distribution can therefore be split across clients so that the server sees only the

noisy sum [16], [17]. Federated marginal synthesizers already exist [18], [19], but they have been evaluated on general tabular benchmarks rather than on downstream detection of rare attacks in IoT traffic, and not against a federated deep generator under the same protections.

This paper studies that comparison on MQTT traffic with PP-FedData, a privacy-preserving federated data generation framework. It addresses four research questions:

- **RQ1:** How much IDS utility does federated synthetic data retain when raw traffic cannot be pooled, and does it add utility when real data are available?
- **RQ2:** Under the same (ε, δ) budget, how does a federated marginal generator compare with a federated CVAE trained with DP-SGD in utility, rare-class recall and communication?
- **RQ3:** What does each protection (DP, SecAgg$^+$, both) cost, and what does it protect, as a function of the protected unit (packet, TCP stream, capture) and of the number of honest gateways?
- **RQ4:** Which configuration should a deployment choose given its trust, privacy and bandwidth requirements?

The contributions are the following. We do not claim a new synthesis algorithm: FedDP-Marginal adapts tree-structured marginal synthesis [14], [20], and federated marginal synthesis has been studied before [18], [19]. What we add is the combination and the evidence.

1) **FedDP-Marginal for IoT-MQTT traffic.** A three-release protocol in which gateways add shares of integer Skellam noise to their count tables and sum them with a real SecAgg$^+$ implementation in Flower; the privacy loss is accounted exactly and reported for every number of honest gateways, with options for a threshold of honest gateways, local DP, and stream- or capture-level protection.
2) **A controlled benchmark against a federated CVAE.** Both generators run on the same non-IID partitions, privacy budgets, secure aggregation and evaluation, which separates the cost of DP, the cost of SecAgg$^+$ and the effect of the generator family.
3) **A requirement-driven configuration guide.** A transparent rule filters the measured configurations by trust, privacy budget, bandwidth and attack priority and selects from their Pareto front.

Section II reviews related work. Section III describes the system model, FedDP-Marginal, its privacy accounting and the CVAE baseline. Section IV gives the experimental setup, Section V reports and discusses the results, and Section VI concludes.

II. Related Work

A. Privacy Protection in Federated Learning

Surveys of privacy-preserving FL for IoT intrusion detection group the techniques into differential privacy, homomorphic encryption, secure multi-party computation, trusted execution environments and blockchain [4]. The last three require heavy computation, special hardware or consensus overhead on constrained gateways, so this work uses DP and secure aggregation.

Differential privacy limits how much a released output can change when one record is added or removed [7]. In the central model a trusted curator adds the noise; in the local model each client perturbs its own contribution and trusts nobody, at a much higher noise level. DP-SGD clips per-example gradients and adds Gaussian noise at every step [12]; its cumulative loss is tracked with Rényi DP (RDP) [21].

Secure aggregation lets the server learn only the sum of client vectors. The protocol of Bonawitz et al. uses pairwise masks and secret sharing to tolerate dropouts [5]; SecAgg$^+$ replaces the complete communication graph with a sparse random graph and reaches polylogarithmic overhead [6]. Secure aggregation adds no statistical noise, but it protects only the individual updates, not the aggregate.

Distributed DP combines both: each client adds a share of the noise and secure aggregation reveals only the sum, which then carries the full calibrated noise. Because secure aggregation works on integers modulo 2^b, discrete mechanisms are needed. Kairouz et al. proposed the distributed discrete Gaussian [16]. Agarwal et al. showed that the Skellam distribution is closed under summation and gave a tight RDP bound for it [17]. Both works train models with many rounds; a release made of a few sums, such as count tables, is a simpler fit.

B. Synthetic Data for Intrusion Detection

Oversampling such as SMOTE interpolates between minority neighbors [22], which requires the minority records to be pooled and is therefore unavailable when a gateway holds few or no attack records. A systematic review of generative AI for IoT intrusion detection found that most studies use generative adversarial networks (GANs) or variational autoencoders for data augmentation and class rebalancing, and noted mode collapse and training instability as recurring problems [8].

Federated generative IDSs follow this line. Hamouda et al. trained a federated conditional GAN on Edge-IIoTset and applied DP-SGD to the downstream classifier rather than to the generator; they identified cryptographic protection of the models as an open challenge [9]. Yao et al. protected the downlink broadcast of synthetic samples with dynamic (t, n)-threshold secret sharing, while the uplink gradients relied on Gaussian DP [10]. Shettar et al. reported a detection F1-score of 96.8% with a central aggregator under DP at $\varepsilon = 4.8$ [11]. None of these works applies secure aggregation to the generator updates or measures how DP and secure aggregation each affect rare-class detection.

C. Marginal-Based Private Synthesis

Marginal-based synthesizers avoid backpropagation altogether. PrivBayes fits a Bayesian network to noisy low-order marginals [13]; MST measures noisy two-way marginals along a maximum spanning tree and post-processes them with a graphical model [14]; AIM selects marginals adaptively and iteratively [15]. Two works move this family to distributed

TABLE I
COMPARISON WITH THE CLOSEST RELATED WORK

Work	FL	IoT	Syn.	DP	SA	Marg.	D-DP	IDS
Hamouda et al. [9]	✓	✓	✓	✓	✗	✗	✗	✓
Yao et al. [10]	✓	✓	✓	✓	✓	✗	✗	✓
Shettar et al. [11]	✓	✓	✓	✓	✗	✗	✗	✓
Kairouz et al. [16]	✓	✗	✗	✓	✓	✗	✓	✗
Agarwal et al. [17]	✓	✗	✗	✓	✓	✗	✓	✗
MST [14]	✗	✗	✓	✓	✗	✓	✗	✗
FLAIM [18]	✓	✗	✓	✓	✓	✓	✗	✗
CaPS [19]	✗	✗	✓	✓	✓	✓	✗	✗
PP-FedData	✓	✓	✓	✓	✓	✓	✓	✓

Syn.: synthetic data generation; SA: secure aggregation or MPC; Marg.: marginal-based DP synthesis; D-DP: distributed DP; IDS: downstream intrusion detection evaluated.

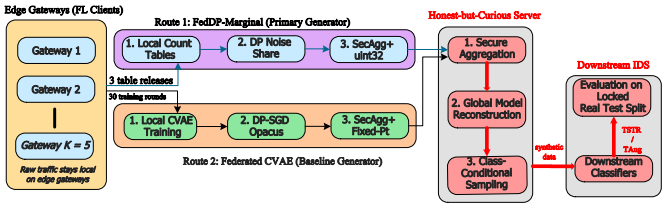


Fig. 1. PP-FedData architecture. Gateways keep raw traffic; FedDP-Marginal sends three releases of noisy count tables and the federated CVAE sends model updates, both through SecAgg$^+$ to an honest-but-curious server, which samples synthetic packets for the downstream IDS.

data. FLAIM adapts AIM to federated clients under zero-concentrated DP and compares it with a secure multi-party variant [18]. CaPS performs marginal selection and measurement inside secure multi-party computation ("DP-in-MPC") [19]. Both were evaluated mainly on general-purpose tabular benchmarks from demographic, financial and biomedical domains; FLAIM also used the legacy DARPA intrusion detection data, but measured workload error and negative log-likelihood rather than downstream rare-class detection. The family has known limits: only low-order dependencies are modeled, continuous attributes must be discretized, and sparse cells have a low signal-to-noise ratio under DP.

D. Research Gap

Table I summarizes the closest works. Based on a non-systematic search of the literature, and to the best of our knowledge, no prior study (i) evaluates a federated marginal generator on IoT-MQTT traffic for downstream rare-class detection, (ii) compares it with a federated deep generator under identical partitions, budgets and secure aggregation, or (iii) combines distributed discrete DP with a real secure aggregation implementation and reports the guarantee as a function of the number of honest clients.

III. SYSTEM MODEL AND METHODOLOGY

A. Architecture and Threat Model

Fig. 1 shows the three parties of PP-FedData. A federation of K edge gateways $\mathcal{K} = \{1, \ldots, K\}$ observes MQTT traffic; gateway k keeps its packets $\mathcal{D}_k$ locally. An aggregation server coordinates the rounds, reconstructs a global generator from the aggregated releases and samples synthetic packets for every class. A downstream IDS is trained on the synthetic packets, alone or with real ones, and is evaluated on real traffic. Two generator routes share this architecture: FedDP-Marginal, which exchanges count tables in three releases, and a federated CVAE, which exchanges model weights over tens of FedAvg rounds.

To model non-IID gateways, the training data are partitioned with the label-skew scheme of Hsu et al. [23]: for each class $c \in \{1, \ldots, C\}$ the shares of the K gateways are drawn as

$$\mathbf{p}_c = (p_{c,1}, \ldots, p_{c,K}) \sim \mathrm{Dir}(\alpha_{\mathrm{dir}} \mathbf{1}_K), \tag{1}$$

and each record of class c goes to gateway k with probability $p_{c,k}$. With $K = 5$ and $\alpha_{\mathrm{dir}} = 0.5$, some gateways receive few or no packets of a rare attack class.

Threat model. The server is honest-but-curious: it follows the protocol but inspects everything it receives. Gateways follow the protocol; poisoning and Byzantine behavior are out of scope. The two protections answer different threats. SecAgg$^+$ [6] hides each gateway's contribution from the server, which learns only the sum. DP bounds what the released sum, and every model or synthetic record derived from it, reveals about one record [7]. By default the protected record is one packet; Section III-C extends the unit to a TCP stream or a capture. Class labels, class quotas, the gateways' sample counts and the preprocessing statistics are treated as public side information outside ε.

B. FedDP-Marginal

FedDP-Marginal adapts tree-structured marginal synthesis [14], [20] to a federation. It models $d = 37$ attributes: 13 numeric attributes, discretized into $B_{\mathrm{fine}} = 64$ equal bins on the standardized range $[-5, 5]$; 19 binary attributes (7 protocol flags and 12 missingness indicators); and 5 categorical attributes (message type, two QoS fields, requested QoS and protocol). Every table is a sum over records, so each release is a single additive query. The protocol has three releases (Fig. 2):

1) **Release 1, fine one-way tables per class.** Each gateway counts every attribute per class on fine bins. From the noisy sum the server places B_{coarse} coarse bins at the quantiles of each numeric attribute and broadcasts the cut points.
2) **Release 2, coarse two-way tables pooled over classes.** Gateways count all $\binom{37}{2} = 666$ attribute pairs on coarse values. The server estimates the mutual information $\hat{I}$ of every pair from the noisy tables (counts clipped at zero and smoothed) and builds a single Chow–Liu tree [20]:

$$T^\star = \arg\max_{T} \sum_{(i,j)\in T} \hat{I}(X_i; X_j), \tag{2}$$

with $\hat{I}(X_i; X_j) = \sum_{a,b} \hat{p}(a,b) \log \frac{\hat{p}(a,b)}{\hat{p}(a)\hat{p}(b)}$, and broadcasts $T^\star$.
3) **Release 3, class-conditional tree tables.** Gateways count the (root, y) table and the $(\mathrm{parent}, \mathrm{child}, y)$ table of every tree edge on coarse values.

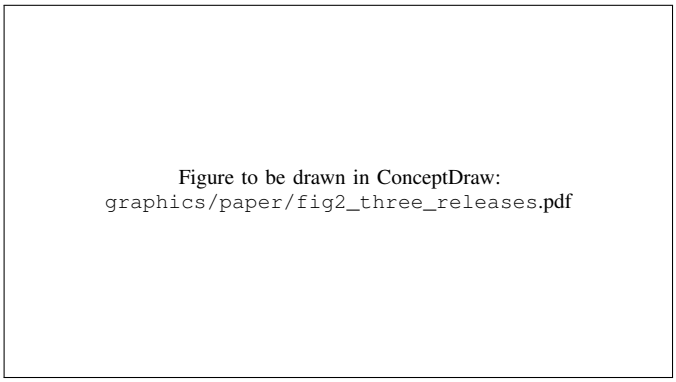


Fig. 2. Three-release protocol of FedDP-Marginal. Each release is a sum of count tables plus Skellam noise shares, computed with SecAgg$^+$ over $\mathbb{Z}_{2^{32}}$. The coarse cut points (after Release 1) and the tree $T^\star$ (after Release 2) are broadcast back before the next release.

Pooling Release 2 over classes is deliberate: a per-class split would divide the signal of each cell by about C while the sensitivity stays the same, which corrupts the tree. Class-specific structure enters through Release 3.

Sampling. For each class c the server samples records along $T^\star$ from the root to the leaves,

$$\hat{P}(\mathbf{x} \mid y=c) = \hat{P}(x_r \mid y=c) \prod_{(u,v)\in T^\star} \hat{P}(x_v \mid x_u, y=c), \tag{3}$$

where r is a fixed root attribute. A sampled coarse value of a numeric attribute is refined to a fine bin with the class's fine histogram from Release 1 and then drawn uniformly inside it. The records are mapped back to the feature space by inverting the preprocessing.

Post-processing (MGr). The refined variant combines every independent noisy estimate of an attribute's class-conditional marginal (from Release 1 and from the incident tree tables) by inverse-variance weighting, clips negative counts, and adjusts each edge table to the fused marginals with iterative proportional fitting. Because it only transforms released values, it costs no privacy budget [7].

C. Privacy Accounting

Sensitivity. Two datasets are neighbors if they differ by one packet. A packet adds one count to one cell of every table it touches, so a release of T_r tables (37, 666 and 37 for the three releases) has ℓ_1 and ℓ_2 sensitivities $\Delta_{1,r} = T_r$ and $\Delta_{2,r} = \sqrt{T_r}$. To protect a unit of several packets (a TCP stream or a capture), each unit is assigned to a single gateway, which uses at most m of its packets. The sensitivities then become

$$\Delta_{1,r} = m\,T_r, \qquad \Delta_{2,r} = m\sqrt{T_r}. \tag{4}$$

Skellam mechanism. The noise of each cell follows the Skellam distribution $\mathrm{Sk}(\mu) = \mathrm{Pois}(\mu/2) - \mathrm{Pois}(\mu/2)$, which has zero mean and variance μ and is closed under summation [17]. For an integer order $\lambda > 1$, a release with variance μ_r per cell satisfies the RDP bound of Agarwal et al. [17, Cor. 3.6]:

$$\varepsilon_r(\lambda) \le \frac{\lambda\Delta_{2,r}^2}{2\mu_r} + \min\left(\frac{(2\lambda-1)\Delta_{2,r}^2 + 6\Delta_{1,r}}{4\mu_r^2}, \frac{3\Delta_{1,r}}{2\mu_r}\right). \tag{5}$$

RDP composes additively over the three releases [21], and the total is converted to (ε, δ)-DP with the conversion used by the Opacus accountant [24]:

$$\varepsilon = \min_{\lambda} \Big[\sum_{r=1}^{3} \varepsilon_r(\lambda) - \frac{\ln\delta + \ln\lambda}{\lambda - 1} + \ln\frac{\lambda-1}{\lambda} \Big], \tag{6}$$

over $\lambda \in \{2, \ldots, 63\} \cup \{64, 72, \ldots, 1024\}$. The budget is split over the releases by a vector $\mathbf{s} = (s_1, s_2, s_3)$ through $\mu_r = \Delta_{2,r}^2/(2 s_r \kappa)$, where κ is the largest value, found by bisection, for which (6) meets the target ε. The split and B_{coarse} are chosen on validation data from $\mathbf{s} \in \{(0.3, 0.1, 0.6), (0.5, 0.1, 0.4), (0.15, 0.05, 0.8)\}$ and $B_{\mathrm{coarse}} \in \{4, 8, 16, 32\}$; the selected settings are $(0.3, 0.1, 0.6)$ with 32 bins at $\varepsilon = 1$, and $(0.15, 0.05, 0.8)$ with 4 and 8 bins at $\varepsilon = 5$ and 10.

Noise sharing and trust. Gateway k adds integer noise $\xi_k \sim \mathrm{Sk}(\mu_r/t)$ to every cell. With $t = K$ (the default, MGs and MGr) the full variance is reached when all gateways are honest. If only h gateways add their share, the aggregate variance is $h\mu_r/t$, and the guarantee is recomputed from (5)–(6) and reported for every h. A threshold $t < K$ keeps the target ε whenever at least t gateways are honest, at the price of more noise when all are. In the local variant (MGl) every gateway adds the full variance μ_r and trusts nobody, so the sum carries $K\mu_r$.

Secure transport. The noisy integer tables are summed with Flower's SecAgg$^+$ [6], [25] over $\mathbb{Z}_{2^{32}}$:

$$S = \sum_{k\in\mathcal{K}} (\mathbf{x}_k + \boldsymbol{\xi}_k) \bmod 2^{32}, \tag{7}$$

where $\mathbf{x}_k$ are gateway k's counts; the server lifts every entry $S \ge 2^{31}$ to $S - 2^{32}$ to recover negative values. Since the counts are at most $N = 88{,}500$ and the noise is small compared with 2^{31}, the lifted sum equals the exact noisy sum: secure aggregation adds no rounding error and receives no privacy credit. Because each release is a single sum, splitting the noise across gateways is valid; it is not valid for the multi-step DP-SGD of the CVAE route.

D. Federated CVAE Baseline

The deep baseline is a label-conditional CVAE [26] on the $D = 60$ encoded features (latent size 16, no batch normalization, so that per-example gradients are independent). It is trained with FedAvg [2] for 30 rounds of 2 local epochs (B3). The DP variant (M1) runs DP-SGD with Opacus 1.6.0 [12],

[24]: per-example gradients are clipped and noised at every local step,

$$\tilde{g} = \frac{1}{B}\Big(\sum_{i=1}^{B} \frac{g_i}{\max(1, \|g_i\|_2/C)} + \mathcal{N}(0, \sigma_k^2 C^2 \mathbf{I})\Big), \quad (8)$$

where g_i is the gradient of example i in a minibatch of size B, C is the clipping norm, and the noise multiplier σ_k is calibrated for each gateway; the federation reports the worst case $\varepsilon = \max_k \varepsilon_k$. M2 aggregates the weights with SecAgg$^+$ (5 shares, reconstruction threshold 3, clipping range 16, 2^{22} quantization levels), and M3 combines M1 and M2. Even with SecAgg$^+$, the CVAE route remains local DP. Synthetic records are drawn from the decoder with $z \sim \mathcal{N}(0, \mathbf{I})$; the DP results use the plain decoder, which depends only on the DP-trained weights. The study first planned the CVAE as its generator. Under DP its utility fell sharply even after per-ε re-tuning at full federated scale (M1o, M3o), which motivated FedDP-Marginal; the CVAE is kept as the baseline.

IV. Experimental Setup

A. Dataset and Preprocessing

We use the DoS/DDoS-MQTT-IoT dataset [1]: 59,624,826 packets, each described by 33 columns, with normal traffic and the DoS and DDoS variants of five MQTT attack families. The attack rows are the authors' CSV exports. The release provides normal traffic only as PCAP files, so we exported every normal packet with TShark into the same columns; applying the same export to the attack captures reproduces the authors' rows once their display filter and two-pass dissection are enabled. The DoS and DDoS variants of each family are merged into one class, so that the IDS detects the attack family. Separating the variants is hard at the packet level: a random forest on real data reaches a macro-F1 of only 0.326 on the eleven sub-classes, and the two variants of basic connect flooding are close to chance. Merging gives six classes (Table II).

Identifiers and capture metadata (frame number, absolute and relative timestamps, source address, free-text information) are removed to avoid leakage [27]. Numeric features with skewness above 2 are log-transformed, all numeric features are standardized and clipped at $\pm 5\sigma$, and categorical fields are one-hot encoded, giving $D = 60$ features. The preprocessing statistics are fitted on the pooled training split, which is a centralized step outside ε. Diagnostic checks with a random forest on the validation split found no shortcut feature: no single column reaches a macro-F1 above 0.9, the missingness indicators alone reach 0.10 (chance 0.17), and a random row split scores only 0.0075 above a split by capture group.

Whole capture groups are assigned to the training, validation or test split. Eight of the eleven DoS/DDoS sub-classes come from a single capture file, so their validation and test rows are contiguous blocks of that capture, separated by 1,000-row buffers; TCP streams crossing a block boundary are dropped. Rows are then sampled to fixed quotas (Table II). The training quotas impose a 60:1 imbalance between `NORMAL` and `WILL`. The validation and test sets are class-balanced, so that the metrics are not dominated by the majority class. The test set (11 capture groups, 20,343 TCP streams) is used only once for the final evaluation. We call `DELAYED`, `SYN`, `INVALID` and `WILL` the rare classes.

TABLE II
Classes, Raw Packet Counts and Split Quotas

Class	Raw packets	Share	Train	Val.	Test
NORMAL	45,580,102	76.44%	60,000	2,000	5,000
BCF	12,280,874	20.60%	20,000	2,000	5,000
DELAYED	601,542	1.01%	3,000	2,000	5,000
SYN	601,439	1.01%	3,000	2,000	5,000
INVALID	290,340	0.49%	1,500	2,000	5,000
WILL	270,529	0.45%	1,000	2,000	5,000
Total	59,624,826	100%	88,500	12,000	30,000

BCF: basic connect flooding; DELAYED: delayed connect flooding; SYN: SYN TCP flooding; INVALID: invalid subscription flooding; WILL: connect flooding with WILL payload.

TABLE III
Evaluated Configurations

ID	Generator	Protection	Rounds
B0, B1a, B1b	none	pooled real data; B1a class weights, B1b SMOTE [22]	–
B3	CVAE	none (FedAvg)	30
M1-eps{1,5,10}	CVAE	DP-SGD at each gateway (local DP)	30
M2	CVAE	SecAgg$^+$	30
M3-eps5	CVAE	DP-SGD and SecAgg$^+$	30
M1o, M3o	CVAE	as M1, M3, re-tuned per ε at full scale	30–50
MGp-, MGs-nodp	Marginal	none; SecAgg$^+$ without noise	3
MGs-eps{1,5,10}	Marginal	distributed Skellam DP and SecAgg$^+$	3
MGr-eps{1,5,10}	Marginal	MGs with post-processing (no extra ε)	3
MGl-eps{1,5,10}	Marginal	local DP, plain sum	3

Suffixes: `-t`t noise share $1/t$ (holds with t honest gateways); `-strm`m and `-cap`m protect a TCP stream or a capture with at most m packets each.

B. Configurations

Table III lists the configurations. The real-data references B0, B1a and B1b pool the raw training data and are therefore not available in a federation. All federated configurations use $K = 5$ gateways and the partition of (1) with $\alpha_{\text{dir}} = 0.5$, drawn once per seed and shared by every configuration. Privacy budgets are $\varepsilon \in \{1, 5, 10\}$ with $\delta = 10^{-5}$ at the packet level. The headline FedDP-Marginal configurations are MG-eps1 = MGr-eps1, MG-eps5 = MGs-eps5 and MG-eps10 = MGs-eps10, the variants selected on validation data. Hyperparameters were tuned on the validation split only (tree-structured Parzen search for the CVAE, a grid for FedDP-Marginal); this selection is not covered by ε.

C. Evaluation Protocols and Metrics

All protocols are scored on the same real test set. *TRTR* trains the IDS on the pooled real training data (B0, B1a, B1b). *TSTR* [28] trains it on 5,000 synthetic rows per class only, which measures what a federation obtains when raw data cannot be pooled. *TAug* adds synthetic rows to the pooled real data until each class has at least 20,000 rows. *Local augmentation* lets each gateway train its own IDS on its local rows, topped up with synthetic rows for every class of which it holds fewer than 5,000 rows (5 gateways $\times$ 3 seeds $= 15$ models). The IDS is

a random forest with 200 trees or a multilayer perceptron with hidden layers (128, 64); for each protocol the classifier with the higher validation macro-F1 is reported.

Utility is measured by macro-F1 over the six classes (primary metric), binary F1 (normal versus attack) and rare-class recall (mean recall of the four rare classes). Fidelity is measured by the Wasserstein-1 distance, the Jensen–Shannon divergence, a correlation distance, and a classifier two-sample test (C2ST) whose AUC is 0.5 when synthetic and real rows are indistinguishable [29]. Privacy is assessed with the formal ε, recomputed independently, and with three membership inference attacks (MIAs): one on the synthetic data, one with access to the released model and calibrated with a reference model [30], and one on the released tables of FedDP-Marginal. Each attack is checked against a positive control on which it must succeed. Communication is the payload measured on the Flower message grid, reported per round and per run.

D. Decision Rule and Environment

Every configuration is run with seeds $\{0, 1, 2\}$, which govern the partition, initialization and sampling. A configuration A is declared better or worse than B only if

$$0 \notin \mathrm{CI}_{95\%}(\bar{\Delta}) \quad \text{and} \quad |\bar{\Delta}| > \max(\mathrm{sd}_A, \mathrm{sd}_B), \tag{9}$$

where $\bar{\Delta}$ is the mean difference in macro-F1 over seeds, the interval comes from a paired bootstrap of the test rows (1,000 class-stratified resamples, identical for all runs), and sd is the seed standard deviation. Otherwise the difference is reported as no effect. Because packets of a TCP stream are correlated, every verdict is repeated with a bootstrap over whole streams. We report no p-values. The federation is simulated with Flower 1.39 [25] and Ray actors on a single Windows 11 machine (CPU only, 16.9 GB RAM), with PyTorch 2.14, Opacus 1.6 and scikit-learn 1.9. Times therefore reflect computation, not network latency. Every run is recorded in a ledger with its configuration hash and code version, and a from-scratch re-run of the seed-0 matrix reproduced all 53 ledger rows exactly.

V. Results and Discussion

A. RQ1: Utility of Federated Synthetic Data

Packet-level detection of these attacks is hard even with all real data pooled. A random forest trained on the imbalanced real training set (B0) reaches a test macro-F1 of 0.447 and a rare-class recall of 0.276; class weights (B1a) and SMOTE (B1b) raise the macro-F1 to 0.492 and 0.504. These references require pooling the raw traffic.

Synthetic data alone (TSTR). Without DP, an IDS trained only on FedDP-Marginal data reaches 0.444 ± 0.017, and one trained on the federated CVAE data (B3) reaches 0.420 ± 0.014. Synthetic-only training thus comes close to the imbalanced real reference while no raw packet leaves a gateway. It stays below the class-balanced references, so part of its value comes from the class balance of the synthetic set.

Synthetic data added to pooled real data (TAug). Here synthetic data do not help. With the random forest, 1 of 13 generator configurations improves on B0 under rule (9), by at most $+0.002$, against $+0.057$ for SMOTE. With the MLP, 10 of 13 improve, by at most $+0.030$, against $+0.140$ for SMOTE. The recall of the rare classes moves by at most 0.012 for any generator, against $+0.063$ to $+0.102$ for class weights and SMOTE.

Local augmentation. The picture changes when the comparison is the gateway itself. A gateway's own IDS, trained on its non-IID local rows, reaches 0.387 ± 0.053. Topping up the classes it lacks with MG-eps5 data raises this to 0.422 ($+0.036$, 15 of 15 client models better), with MG-eps10 to 0.425 ($+0.039$, 14 of 15), and with MG-eps1 to 0.419 ($+0.033$, 10 of 15). For reference, a classifier trained directly with FedAvg and class weights reaches 0.429, which the rule cannot separate from the CVAE route (0.420). A federation that needs only a detector, and no shareable data, can therefore train the detector directly.

The answer to RQ1 is that federated synthetic data are useful where raw traffic cannot be pooled: to train a shared IDS or to fill in the classes a gateway does not see. They do not improve an IDS whose real training data are already pooled.

TABLE IV
TSTR Utility and Communication at Equal Privacy Budgets

ε	Configuration	Macro-F1	Bin. F1	Rare rec.	MB/rd	MB/run
1	CVAE (M1-eps1)	0.227 ± 0.030	0.805	0.132	1.46	43.7
	MG (MGr-eps1)	$\mathbf{0.389 \pm 0.010}$	0.875	0.253	0.20	0.60
5	CVAE (M1o-eps5)	0.287 ± 0.027	0.800	0.168	0.54	26.8
	MG (MGs-eps5)	$\mathbf{0.408 \pm 0.003}$	0.901	0.293	0.15	0.45
10	CVAE (M1o-eps10)	0.289 ± 0.040	0.850	0.132	1.40	41.9
	MG (MGs-eps10)	$\mathbf{0.429 \pm 0.008}$	0.906	0.331	0.16	0.47
∞	CVAE (B3)	0.420 ± 0.014	0.929	0.367	4.22	126.7
	MG (MGs-nodp)	0.444 ± 0.017	0.908	0.335	0.17	0.52

CVAE rounds: 30 (50 for M1o-eps5); FedDP-Marginal: 3. Real-data references (pooled, TRTR): B0 0.447, B1a 0.492, B1b 0.504.

B. RQ2: FedDP-Marginal Versus the Federated CVAE Under DP

Table IV compares the two routes at equal budgets; for the CVAE we report its best configuration on validation data at each ε, including the versions re-tuned at full scale. FedDP-Marginal leads at every budget: 0.389, 0.408 and 0.429 against 0.227, 0.287 and 0.289. The gaps of 0.12–0.16 are 3.5 to 5.4 times the larger seed standard deviation. Its rare-class recall is 1.7 to 2.5 times as high (0.253–0.331 against 0.132–0.168). It also needs 3 rounds of 0.15–0.20 MB instead of 30–50 rounds of 0.54–1.46 MB, i.e., 0.45–0.60 MB per run against 27–44 MB, a reduction of more than 98% at each budget. Fig. 3 shows how utility varies with ε.

The cost of DP differs between the routes. Measured against its own non-private release with rule (9), FedDP-Marginal loses 0.055 (random forest) and 0.063 (MLP) at $\varepsilon = 1$, 0.036 and 0.043 at $\varepsilon = 5$, and 0.031 and 0.016 at $\varepsilon = 10$; the last difference is within noise. For the CVAE route, 8 of 12 such comparisons are worse, with losses of up to 47% of the synthetic-only macro-F1, and re-tuning its DP hyperparameters

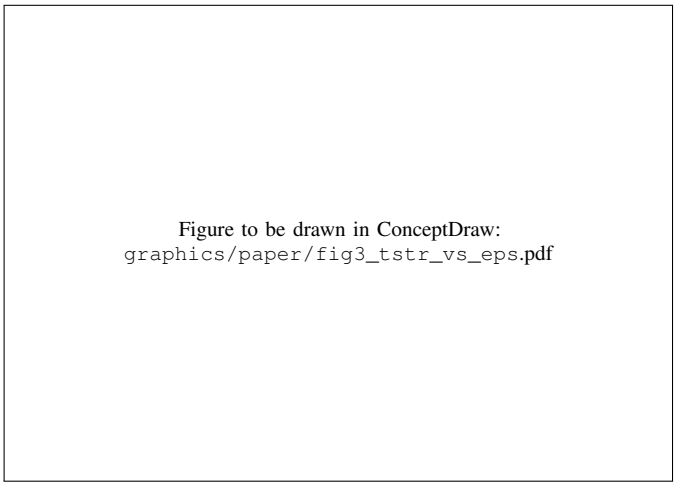


Fig. 3. TSTR macro-F1 (bars: one seed standard deviation) against the privacy budget for FedDP-Marginal with distributed noise (MG) and local noise (MGl), and for the best CVAE configuration. Dashed lines: random forests trained on pooled real data.

per budget at full federated scale did not improve on the baseline beyond the seed spread. Without DP the marginal generator is also better than the CVAE (random forest $+0.048$, MLP $+0.024$). Local noise costs more at small budgets: MGl reaches 0.253, 0.374 and 0.413 at $\varepsilon = 1$, 5 and 10, because the sum carries K times the variance.

In fidelity, FedDP-Marginal matches the per-column distributions more closely (Wasserstein distance 0.047–0.066 against 0.20–0.25 for the DP-trained CVAE). However, a classifier still separates synthetic from real rows almost perfectly for every generator, private or not (C2ST AUC 0.9995–1.0000). Utility for an IDS therefore does not mean that the synthetic packets look real.

C. RQ3: What Each Protection Costs and Protects

Secure aggregation. For FedDP-Marginal, SecAgg$^+$ costs 1.34 times the bytes per round (0.174 against 0.129 MB) and 1.10 times the time of the three rounds (23.1 against 21.1 s, mostly simulator start-up). Utility is unchanged, because the secure sum equals the exact sum. For the CVAE, SecAgg$^+$ costs 1.12 times the time per round and 1.52 times the bytes per parameter, and none of 8 utility comparisons shows an effect. DP-SGD, in contrast, multiplies the time per round by 4.3–4.5.

Empirical leakage. The attack on the released tables of FedDP-Marginal passes its positive control (AUC 0.583). It detects the release without DP (AUC 0.560; true-positive rate 1.6% at 1% false positives) and less the releases with DP (0.514, 0.522 and 0.531 at $\varepsilon = 1$, 5, 10). SecAgg$^+$ alone does not change this number, since it hides each gateway's tables, not what their sum reveals. For the CVAE, the attack on synthetic data stays near chance (0.488–0.498) at every ε, including ∞, and fails its positive control (0.519). The attack with model access passes its control (0.664) but finds no leakage in the federated models (0.496–0.500). At this scale there is therefore no measurable leakage for DP to reduce in the CVAE route; DP provides the formal guarantee.

TABLE V
PROTECTION OPTIONS OF FEDDP-MARGINAL ($\varepsilon \leq 5$)

Config.	Protection	Macro-F1	$\varepsilon_{h=5}$	$\varepsilon_{h=1}$	MIA
MGp-nodp	none	0.444 ± 0.017	∞	∞	0.560
MGs-nodp	SecAgg$^+$ only	0.444 ± 0.017	∞	∞	0.560
MGl-eps5	local DP	0.374 ± 0.010	5.00	5.00	–
MGs-eps5	distributed DP, $t = 5$	0.408 ± 0.003	5.00	14.2	0.522
MGs-t3-eps5	distributed DP, $t = 3$	0.408 ± 0.025	3.72	10.1	0.520
MGs-t2-eps5	distributed DP, $t = 2$	0.396 ± 0.005	2.94	7.6	0.511
MGs-strm2-eps5	stream unit, $m = 2$	0.371 ± 0.004	5.00	13.3	–
MGs-cap25-eps5	capture unit, $m = 25$	0.203 ± 0.037	5.00	13.0	–

ε_h: guarantee when h of the 5 gateways add their noise share; with $t < 5$ the target holds whenever t gateways are honest. MIA: calibrated AUC of the attack on the released tables (positive control 0.583).

Trust and protected unit. Table V summarizes the options of FedDP-Marginal at $\varepsilon = 5$. With the noise split over all five gateways, the guarantee degrades to $\varepsilon = 14.2$ if only one gateway is honest; this value grows with the federation (25.4 for 10 and 49.9 for 20 gateways). A threshold $t = 3$ keeps $\varepsilon \leq 5$ with any three honest gateways and changes the utility by only -0.001 to $+0.003$ across budgets. Protecting a whole TCP stream costs 0.02–0.04 macro-F1, because 90% of the streams contribute a single packet. Protecting a whole capture, of up to 1,277 packets, leaves too little data (0.15–0.20). Because only sums are released, FedDP-Marginal is insensitive to how the rows are spread: with α_{dir} from 0.1 to 10 and 5 to 20 gateways, its macro-F1 at $\varepsilon = 5$ stays within 0.396–0.431.

D. RQ4: Requirement-Driven Configuration

A single utility ranking does not settle the choice: the best configurations differ more in what they protect and cost than in macro-F1. The recommendation rule therefore proceeds in five steps. It (i) keeps only valid configurations (those whose guarantee covers everything they release); (ii) applies the trust requirement (an untrusted server needs SecAgg$^+$ or local DP, and colluding gateways need local DP or a threshold t); (iii) removes configurations that exceed the limits on ε, bandwidth, rounds or MIA AUC; (iv) keeps the Pareto front over utility, ε and cost, where utility within the seed spread, ε within 2% and cost within 5% count as equal; and (v) breaks ties by the priority (macro-F1 or rare-class recall). Table VI lists the outcome for eleven deployment scenarios.

Fig. 4 places every valid configuration by utility and total traffic. All FedDP-Marginal variants stay below 2.5 MB per run, at least an order of magnitude below the CVAE routes (26.8–192 MB), and they make up 9 of the 10 configurations on the Pareto front. A deployment that cannot trust the server but trusts its gateways should take MGs-eps5 or MGs-eps10. If gateways may collude, it needs local noise or a threshold, and the table quantifies the cost of each. A capture-level guarantee is not achievable on this data.

E. Discussion and Limitations

Why the marginal route degrades less under DP. FedDP-Marginal spends its budget on three releases, each a single

TABLE VI
RECOMMENDED CONFIGURATION PER DEPLOYMENT REQUIREMENT

Scenario	Requirement	Choice	F1	ε
lab	all parties trusted, no DP	MGp-nodp	0.444	∞
hidden-tables	untrusted server, no DP	MGr-eps10	0.433	10
cloud-aggregator	untrusted server, $\varepsilon \leq 5$	MGs-eps5	0.408	5
rare-attacks	as above, rare recall first	MGs-eps5	0.408	5
strict-privacy	untrusted server, $\varepsilon \leq 1$	MGr-t3-eps1	0.392	1
low-bandwidth	$\varepsilon \leq 10$, ≤ 100 MB total	MGr-eps10	0.433	10
consortium	gateways may collude, $\varepsilon \leq 10$	MGl-eps10	0.413	10
honest-majority	≥ 3 honest, $\varepsilon \leq 10$, MIA < 0.55	MGs-t3-eps10	0.428	10
gateway-dropout	≥ 3 honest, $\varepsilon \leq 5$	MGs-t2-eps5	0.396	3.94
session-privacy	TCP-stream unit, $\varepsilon \leq 5$	MGs-strm2-eps5	0.371	5
capture-privacy	capture unit, $\varepsilon \leq 10$	MGs-cap25-eps5	0.203	5

F1: TSTR macro-F1; ε: the guarantee that holds under the scenario's trust assumption.

Figure to be drawn in ConceptDraw:
`graphics/paper/fig4_utility_vs_traffic`.pdf

Fig. 4. TSTR macro-F1 against total traffic per run (log scale) for all valid configurations. MG: FedDP-Marginal, including its Gaussian (MGd) and second-order (MGb) ablations. Filled markers: Pareto front over utility, ε and cost.

sum with a known sensitivity, so the noise is added once per cell. Its pooled pairwise release keeps the per-cell signal high. Because each release is a sum, distributed noise and secure aggregation fit it exactly. DP-SGD instead adds noise at every step of every gateway, and its noise cannot be shared across gateways. In return, the marginal route captures only pairwise dependencies through one tree, and second-order structures did not help on this data.

Limitations.

- **Data:** The study uses one dataset at the packet level, and generated packets have no stream context. The normal traffic was re-exported from PCAP files and, unlike the attack rows, includes broker-sent packets (44.2% of the normal training rows). Re-scoring all runs on client-sent test rows changes the macro-F1 by -0.019 on average and keeps the order of the configurations (Spearman 0.993).
- **Privacy scope:** ε protects one packet by default. One capture group contributes up to 188 packets of a rare class, so the group bound for a whole capture is vacuous. Labels, class counts, preprocessing statistics and validation-based tuning lie outside ε.
- **Trust and system:** The study assumes honest-but-curious parties, and the federation is simulated on one machine, so no real network or device costs are measured.
- **Statistics and scope:** Three seeds give a rough spread, and the verdicts hold for this test split. Federated SMOTE, FLAIM or AIM themselves, and other deep generators were not compared.

VI. CONCLUSION

This paper presented PP-FedData, a framework for generating synthetic IoT-MQTT traffic across gateways that cannot share raw packets, and compared two federated generators under the same partitions, privacy budgets and secure aggregation. FedDP-Marginal, which releases three sums of noisy count tables with Skellam noise split across gateways and summed by SecAgg$^+$, kept 0.389–0.429 test macro-F1 at $\varepsilon = 1$–10, against 0.227–0.289 for a federated CVAE trained with DP-SGD, while sending less than 1 MB per run. Secure aggregation cost little and no utility; DP cost FedDP-Marginal 0.02–0.06 macro-F1 and measurably reduced what an attack on its released tables could detect. The results also bound the claim. Synthetic data did not improve an IDS whose real data could be pooled, synthetic packets remained distinguishable from real ones, and the guarantee is per packet unless a stream-level unit is chosen. The framework is therefore a way to train or complete an IDS where traffic must stay local, and its requirement-driven rule tells a deployment which protection to choose and what it will cost.

Future work will re-extract the normal traffic with the dataset authors' export settings, evaluate other IoT datasets and protocols, compare against FLAIM and AIM directly, account for hyperparameter selection privately, and measure the protocols on physical gateways over a real network.

ACKNOWLEDGMENT

This research was conducted within the Capstone Project Program at FPT University Hanoi Campus, Department of Information Technology. The authors would like to express their sincere gratitude to supervisor Dr. Nguyen Anh Nhat and co-supervisor MSc. Luu Minh Tri for their invaluable guidance, domain expertise, and continuous encouragement throughout this study.

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